MIL-T-82169A(MC)
18 March 1969
SUPERSEDING
MIL-T-82169(MC)
30 October 1967

MILITARY SPECIFICATION

TERMINAL, TELEGRAPH-TELEPHONE: TH-85/GCC

1. SCOPE

*1.1 This specification covers requirements for one type terminal, telegraph-telephone for operation of direct current (dc) teleprinters over voice frequency (vf) facilities.

*2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-T-152	- Treatment, Moisture and Fungus-Resistant, of Communication, Electronic and Associated Electrical Equipment.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-P-15024	- Plates, Identification-Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment.
MIL-E-16400	- Electronic Equipment, Naval Ship and Shore, General Specification.
MIL-E-17555	- Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.
MIL-E-21981	- Electronics Type Designation, Identification Plates and Markings, Requirements for.
MIL-R-22732	- Reliability Requirements for Shipboard and Ground Electronic Equipment.

FSC 5805

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STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-188	- Military Communication System Technical Standards.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-826	- Electromagnetic Interference Test Requirements and Test Methods.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

*3.1 General. The Terminal, Telegraph-Telephone, TH-85/GCC, shall consist of a device capable of converting dc binary signals to vf signals for transmission via a radio or wire line path. It shall also be capable of receiving vf signals and converting them to dc binary signals for operation of electromechanical stations. The terminal shall be capable of conversion at all speed rates up to and including 100 words per minute at 7.0 or 7.42 unit code, in half-duplex or full-duplex, 2-wire operation. The capability to utilize standard field telephones over the 2-wire vf loop from the front panel of the converter shall be provided.

*3.1.1 Type and function of equipment. The terminal shall be a lightweight, solid-state, tactical type high performance converter and shall provide half-duplex or full-duplex teleprinter operation over a 2-wire line. The terminal shall also provide the following:

(a) Capability of operation in point-to-point systems, network systems or switched systems.

(b) Compatibility with AN/TGC-14 ( )/V, AN/TGC-15 ( ), Model 28 type, TT-76 ( )/GGC and TT-4 ( )/TG teletypewriters, and TH-5/TG, and TH-22/TG telegraph terminals. For the two latter equipments, full compatibility is provided except for the telegraph break feature.

(c) Low power consumption.

(d) High reliability and ease in maintainability.

*3.1.2 Mount. The telegraph-telephone terminal requires a mount, MT-4076/GGC, for operation; however, this mount is not a part of the Terminal, Telegraph-Telephone; TH-85/GCC.

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*3.2 First article. Prior to submission of any complete production units for acceptance, the contractor shall furnish first articles for testing under the direction and supervision of the cognizant Government inspector to determine conformance to this specification. First article is defined as a production sample which will be identical to the production items which the contractor will subsequently produce in fulfillment of the contract. Unless otherwise specified, a quantity of five first articles shall be furnished (see 6.2). Approval of the first articles by the activity concerned shall not relieve the contractor of the responsibility to supply equipment conforming to this specification. Examination and testing shall be as specified herein. Any changes or deviations of the production units from the first articles shall be subject to the approval of the contracting officer. The items subjected to first article tests may be included in the final shipment as part of the total quantity to be supplied under the contract, provided they have been completely refurbished.

*3.3 Reliability assurance. Reliability shall receive major emphasis during the design, fabrication and construction of terminals. This emphasis shall be reflected in the assembly processes and the determination of operational techniques. The first articles, production and sample items shall be subjected to the mechanical and electrical tests specified in Section 4. Mechanical wear shall be such that mechanical or functional performance is not affected. Abrasion of painted surfaces is allowable. Electrical performance of the terminal shall not be degraded. The contractor is required to demonstrate that the terminal is capable of calculated mean time between failure (MTBF) of not less that 15,000 hours. The contractor is required to demonstrate actual MTBF of not less than 1,000 hours under standard and service conditions. Test level No.2 of MIL-R-22732 shall be utilized in establishing demonstrated test conditions.

*3.4 Design and construction. Design and construction shall be as follows.

*3.4.1 Mechanical parts. Where mechanical parts are employed, whose functions are to rotate, cam or oscillate against other mechanical parts and the best choice of materials includes other than those specified in MIL-E-16400, use of the other materials may take precedence.

*3.4.2 Parts and materials. In the selection of parts and materials, fulfillment of major objectives shall be the prime consideration. In all cases, general material requirements as to flammability, fungus resistance and nontoxicity as specified in MIL-T-152 shall be met.

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*3.4.3 Nonstandard parts. When it has been determined that requirements cannot be met by use of parts and materials specified in MIL-E-16400, the contractor shall immediately inform the contracting officer by submission of detailed data as required by MIL-E-16400 and shall request approval.

*3.4.4 Components. Parts and materials approved by specifications listed in MIL-E-16400 shall be given first consideration.

*3.4.5 Case material. Construction of the case and front panel of the terminal and mounting base shall be made of cast aluminum alloy. Where panels are removable from the case and cables enter the front panel or case, such openings shall be protected with gaskets or packing to assure water-tight protection for the equipment when in transit or in an operating condition.

3.4.6 Treatment and painting. Treatment and painting shall conform to MIL-T-704. Color shall match Marine Corps green No.23 (see 6.3).

*3.4.7 Standard conditions. The following conditions shall be used as the basis for establishing normal requirements for conducting laboratory and bench testing of the equipment:

(a) Room ambient temperature +20° to +30° centigrade (C).

(b) Alititude, normal ground.

(c) Room ambient humidity up to 95 percent relative.

(d) Input voltage 115 alternating current volts (Vac), 60 or 400 Hertz (Hz) ±2 percent, single-phase.

(e) No shock or vibration present.

*3.4.8 Service conditions. The equipment shall operate satisfactorily under any of the following service conditions:

(a) Ambient temperature range -40° C to +65° C.

(b) Altitude, sea level to 10,000 feet.

(c) Ambient humidity range 0 to 95 percent relative.

(d) Vibration (See 4.3).

(e) Shock (See 4.4).

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(f) Input voltage 100-130 Vac 45-65 and 340-460 Hz, single-phase.

(g) Immersion (leakage) (see 4.5).

(h) Salt fog environment MIL-STD-810 (see 4.11).

(i) Blowing sand and dust per MIL-STD-810 (see 4.12).

(j) Temperature shock per MIL-STD-810 (see 4.8).

(k) High temperature per MIL-STD-810 (see 4.6).

(l) Low temperature per MIL-STD-810 (see 4.7).

*3.4.9 Storage conditions. The equipment shall be capable of being stored under the following conditions with a subsequent period of operation:

(a) Ambient temperature range -62°C to +71°C .

(b) Altitude Sea level to 50,000 feet.

*3.4.10 Stability.

*3.4.10.1 Electrical. The equipment shall operate as specified herein for at least 500 hours, either continuously or intermittently, independent of component failure without necessity for readjustment of any controls other than those located on the front panel.

*3.4.10.2 Mechanical. All controls and adjustments shall be designed so that their position of adjustment will not be altered when the equipment is subjected to the specified service conditions.

*3.4.11 Interchangeability. All components and parts shall be physically, functionally, mechanically, and electrically interchangeable with any other equipment produced under this specification.

*3.4.12 Identification and marking. Subassemblies shall be identified with the manufacturer's part number, steel stamped, rubber stamped or silk screened on a fixed surface if available, or on a permanent tag affixed in a manner which shall not impair operation of the assembly, and shall not be easily

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removable. The marking shall be visible when mounted on the assembly.

*3.5 Fungus resistance. The equipment shall be rendered fungus proof by the selection of parts and materials which are nonnutrient for fungus.

*3.6 Electrical characteristics. The electrical characteristics at a line voltage of 115 Vac shall be as specified in the following paragraphs and applicable requirements of MIL-STD-188.

*3.6.1 Teleprinter send, receive and transmit loops. The terminal shall furnish 20 ±2 milliamperes (mA) dc to an external loop resistance of 100 to 1,800 ohms. Open circuit voltage shall be 48 ±10 percent. Both the receive and transmit loops shall be 2-wire.

*3.6.1.1 Copy loop. The teleprinter copy loop shall be 2-wire. It shall provide home copy to a second printer when a single teleprinter is utilized for full-duplex, split function operation.

*3.6.1.2 Type of operation. Type of operation shall be neutral operation, operating either half-duplex or full-duplex.

*3.6.2 Voice frequency loop. The vf loop shall exhibit the following characteristics.

*3.6.2.1 Type of loop. The loop shall be 2-wire, half-duplex or full-duplex.

*3.6.2.2 Type of modulation. The modulation shall be frequency shift. Frequencies are listed as follows and are dependent on the mode selected by the front panel selector on the terminal.

	Transmit frequency Hz		Receive frequency Hz	
Switch Position	Mark	Space	Mark	Space
HDX NOR	1317.5	1232.5	2682.5	2767.5
FDX NOR	1317.5	1232.5	2682.5	2767.5
HDX REV	2682.5	2767.5	1317.5	1232.5
FDX REV	2682.5	2767.5	1317.5	1232.5
TH-5	1317.5	1232.5	1317.5	1232.5

Note: Tolerance for low group frequencies is ±4 Hz. Tolerance for high group frequencies is ±8 Hz.

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*3.6.2.3 Line level. The vf transmitting level shall be -1 to +2 decibels referred to 1 milliwatt (dBm). The vf receive level shall be -40 to +1 dBm.

*3.6.2.4 Loop impedance. The vf loop impedance shall have the following nominal values:

Frequency, Hz	Impedance, ohms
16-25	15,000 minimum
1,000-3,500	600 ±10 percent

*3.6.2.5 Distortion of signal. The total telegraph bias distortion of a transmitting and receiving circuit shall be less than 5 percent.

*3.6.3 Terminal input power requirements. The terminal shall operate over an input range of 100-130 Vac, 45-65 and 340-460 Hz within specified limits of loop electrical characteristics. Power consumption with SEND, RECEIVE and COPY loops marking, shall not exceed 14 watts at 115 Vac, single-phase 60-Hz input.

*3.7 Front panel controls. The front panel of the terminal shall contain the following and shall be identified as follows:

Panel Controls	Switch Position	Function
VF binding posts E71 and E72		Provide connections for the vf loop.
TEL binding posts E73 and E74		Provide connections for a telephone set.
SEND binding posts E75 and E76		Provide dc loop connections for the sending side of the teleprinter.
REC binding posts E77 and E78		Provide dc loop connections for the receiving side of the teleprinter.

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Panel Controls	Switch Position	Function
COPY binding posts E79 and E80		Provide dc loop connections for the receiving side of the teleprinter to provide home copy in full-duplex operation.
GRD binding post E81		For grounding the terminal.
Indicator lamp DS-7	TH-5 HDX & FDX	Indicates when circuit is ready and operator may start sending. Indicates line failure, low received level, or failure of terminal.
Selector-switch S71 (12-5 position rotary).	TH-5	Arranges terminal for operation to a TH-5/TG or TH-22/TG.
	HDX REV	Arranges terminal for half-duplex operation to another TH-85/GCC switched HDX NOR.
	HDX NOR	Arranges terminal for half-duplex operation to another TH-85/GCC switched HDX REV or over a TH-81/GCC system to another TH-85/GCC switched HDX NOR.
	FDX REV	Arranges terminal for full-duplex operation to another TH-85/GCC switched FDX NOR.

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Panel controls	Switch Position	Function
	FDX NOR	Arranges terminal for full-duplex operation to another TH-85/GCC switched FDX REV or over a TH-81/GCC system to another.
Power switch S72, two position toggle	On Off	Switches on ac power Switches off ac power.
Fuse F71 (1 ¼ ampere)		Protects ac supply.

*3.8 Dimensions. The maximum external dimensions of the terminal, shall not exceed 9 inches wide by 6-3/4 inches deep by 3-1/4 inches high.

*3.9 Weight. The weight of the terminal shall not exceed 8 pounds.

*3.10 Identification plates. Identification plates shall be in accordance with MIL-P-15024 and shall be of aluminum alloy where possible. Drawings for identification plates shall be submitted for approval 90 days prior to first article tests. Nomenclature shall be requested in accordance with MIL-E-21981 90 days prior to first article testing.

*3.11 Workmanship. Terminals covered by this specification shall be manufactured and finished in accordance with MIL-E-16400 for Marine Corps green, color No.23 (see 6.3).

*4 QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies

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and services conform to prescribed requirements.

*4.2 Inspection and tests. Inspection and tests shall consist of all tests specified herein to demonstrate that the terminals submitted for approval meet all requirements of this specification. First article testing will be witnessed by a representative of the contracting officer.

*4.2.1 Tests. The tests required for first article, production and sample tests are listed in Table I.

4.2.1.1 Inspection and test procedures. The detailed procedures for conducting tests shall be prepared by the contractor and submitted to the contracting officer for approval 30 days prior to the initiation of any official testing.

*4.2.1.2 Tests to be performed. Terminals shall be tested as follows in the quantities indicated below:

(a) First article. Unless otherwise specified, a quantity of five complete units shall be used to complete these tests (see 3.2). Tests performed on one unit need not be repeated on another. The Government representative will select which terminal will be subjected to each test.

(b) Production. All terminals shall be subjected to the production tests.

(c) Samples. Sample terminals shall be randomly selected by the cognizant Government inspector.

Qty. produced	Qty. to be tested
1-49	1
50-99	2
100-149	3
150-199	4
200-300	5
	One additional unit per 100 units

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Table I Tests

ELECTRICAL	PARA	ENVIRONMENTAL	PARA
Teleprinter loops	3.6.1	Standard Conditions	3.4.7
VF loop transmit	3.6.2	Vibration	4.3
VF loop receive	3.6.2	Shock	4.4
Loop impedance	3.6.2.4	Immersion	4.5
Power consumption	3.6.3	High temp.	4.6
Distortion	.6.2.5	Low temp.	4.7
Interference	4.13	Low pressure	4.10
		Salt fog	4.11
		Sand and dust	4.12
		Temp. shock	4.8
		Humidity	4.9

Teleprinter loops	VF loop transmit	VF loop receive	Loop impedance	Power consumption	Distortion	Interference	Standard Conditions	Vibration	Shock	Immersion	High temp.	Low temp.	Low pressure	Salt fog	Sand and dust	Temp. shock	Humidity
FA P	FA P	FA P	FA P	FA P	FA P	FA	FA P	FA	FA	FA	FA S	FA S	FA	FA	FA	FA	FA
FA	FA	FA		FA					FA	FA	FA S	FA S	FA	FA	FA	FA	FA
FA	FA	FA		FA					FA	FA	FA S	FA S	FA	FA	FA	FA	FA
FA	FA	FA		FA					FA	FA	FA S	FA S	FA	FA	FA	FA	FA
FA	FA	FA		FA					FA	FA	FA S	FA S	FA	FA	FA	FA	FA
FA	FA	FA		FA					FA	FA	FA S	FA S	FA	FA	FA	FA	FA
FA	FA	FA		FA					FA	FA	FA S	FA S	FA	FA	FA	FA	FA
FA	FA	FA		FA					FA	FA	FA S	FA S	FA	FA	FA	FA	FA
FA	FA	FA		FA					FA	FA	FA S	FA S	FA	FA	FA	FA	FA
FA	FA	FA		FA					FA	FA	FA S	FA S	FA	FA	FA	FA	FA

NOTE: FA = First article test S= Sample test P= Production test

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*4.3 Vibration test. One unit shall be subjected to the following vibration test with the unit operating. The unit shall be affixed to the vibration table and be subjected to the following conditions. Vibration in the frequency range of 10 to 55 Hz at 5 gravity (g) or 0.060 inch total excursion, whichever is the limiting value. The frequency change shall be at a rate of less than 1 Hz. The duration of the test shall be 1/2 hour in each of the three planes. There shall be no rigid body resonances and no signs of internal or external physical deterioration.

*4.4 Shock test. One unit shall be subjected to the following shock test with the unit operating. The unit shall be mounted rigidly on the adapter plate of the LAB package testing machine. The machine shafts shall be operated in phase and adjusted in speed to produce average peak accelerations of 10 g's with occasional peaks of 20 g's as measured on the case of the equipment. Malfunction is permissible provided that normal operation can be restored by adjustment of front panel controls. There shall be no signs of internal or external physical deterioration.

*4.5 Immersion (leakage) test. The unit shall be submerged in water to a depth of 3 feet, measured from its highest point. The unit shall be held in this position for 1 hour, and during the test all panel controls shall be operated three times through their full travel. There shall be no entry of water into the unit and no signs of internal or external physical deterioration.

*4.6 High temperature test. The unit shall be tested in accordance with procedure I, method 501, of MIL-STD-810. The unit shall be operated at a temperature not less than 60° C nor greater than 65° C.

*4.7 Low temperature test. The unit shall be tested in accordance with procedure I, method 502, of MIL-STD-810. The unit shall be operated at a temperature not greater than -40° C.

*4.8 Temperature shock test. The unit shall be tested in accordance with procedure I, method 503, of MIL-STD-810 with the exception that the maximum temperature shall be 71° C.

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*4.9 Humidity test. The unit shall be tested in accordance with procedure I, method 507, of MIL-STD-810.

*4.10 Low pressure test. The unit shall be tested in accordance with procedure I, method 500, of MIL-STD-810.

*4.11 Salt fog test. The unit shall be tested in accordance with procedure I, method 509, of MIL-STD-810. Salt deposits shall be removed by rinsing with tap water.

*4.12 Sand and dust test. The unit shall be tested in accordance with procedure I method 510 of MIL-STD-810.

*4.13 Interference test. The equipment shall be tested for conducted interference and radiated interference, in accordance with MIL-STD-826.

*4.14 Test reports. The contractor shall prepare and submit to the procuring activity copies of all test reports not less than 10 days after completion of tests.

*4.15 Approval. Approval of the first articles shall be by the contracting officer upon satisfactory completion of all tests.

*4.16 Acceptance. With exception of the first articles, acceptance of all items shall be by the cognizant Government inspector.

*5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. Preservation, packaging and packing shall be level A or C in accordance with MIL-E-17555 as specified in the contract or order (see 6.2).

5.2 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

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6. NOTES

6.1 Intended use. This equipment is to be employed with tactical teleprinter equipment and will furnish half-duplex or full-duplex 2-wire operation, under tactical conditions.

*6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Quantity of first articles required (see 3.2 and 4.2.1.2 (a)).

(c) Level of preservation, packaging, and packing (see 5.1).

6.3 Color. Color chips for Marine Corps green, No. 23 may be obtained from the Commandant of the Marine Corps, (Code CSY-10), Headquarters, U.S. Marine Corps, Washington, D. C. 20380.

*6.4 First article. The first articles shall be examined and tested for approval, at the contractor's plant, or at an independent commercial testing laboratory acceptable to the procuring activity. First article tests shall be witnessed by a Government representative of the procuring activity. Approval of the first articles shall be by the contracting officer.

*6.5 Marginal notation. The margins of this specification are marked with an asterisk to indicate where changes, (additions, modifications, corrections, deletions), from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Preparing activity:
Navy - MC
Project No. 5805-N134

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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minimum of 24 hours operation under the conditions of the

accumulated